

I have read and evaluated the manuscript “*There is no single density: star-forming regions and galaxies hold more dense ionized gas than long assumed*” by Mendez-Delgado et al. The manuscript presents an interesting analysis of nebular electron density diagnostics and argues that the systemic differences among densities inferred from different emission line ratios arise from the different each diagnostics sensitivity to different ranges of a broad density distribution. The general idea is interesting and potentially important, however, as written, I do not think several of the manuscript’s main conclusions are established by the data and modeling presented. In particular, the manuscript transitions from trying to demonstrate that density-selection effects can occur to arguing that they are the primary explanation for the observed hierarchy in low- and high-ionization density diagnostics without robust demonstration. There are also issue with the interpretation of the proposed maximum-sensitivity density variable  $n_M$ , the heterogeneous robustness of the density diagnostics used to establish the relation reported, and the degree to which the forward modeling discriminates between density-selection effects and genuine ionization-dependent density structure. My main concerns are described below. In most cases I think there are concrete analyses that could substantially strengthen the paper and clarify which conclusions are supported by the present data and which need further investigation.

## Major concerns

### 1. The manuscript should more carefully distinguish what is demonstrated about density structure from what is demonstrated about ionization stratification

There are three related claims in the manuscript that should be separated more clearly.

- I. *Claim: HII regions contain density structure.* This claim is very likely true and has a substantial literature behind it (e.g., Osterbrock & Flather 1959; Rubin 1989; Copetti+2000) and more recent work has specifically demonstrated diagnostic-dependent and ionization-dependent density structure in the extreme star-forming systems most relevant to the present manuscript (e.g., Mingozi+2022; Isobe+2023; Méndez-Delgado+2023; Topping+2025; Martinez+2025). The manuscript should place this result more explicitly in the context of previous work on density inhomogeneities rather than presenting the existence of density structure itself as a novel conclusion.
- II. *Claim: Even within the same ionic zone, diagnostics with different critical-density response functions can return different emissivity-weighted densities.* The Orion data presented here provide interesting tentative support for this, but I would describe the present evidence as suggestive rather than definitive. In particular, the traditional, relatively clean density doublets yield densities that are largely consistent within the uncertainties, while much of the larger dynamic range arises from less straightforward transauroral and Fe diagnostics discussed below.
- III. *Claim: The systematic density increase seen from low- to high-ionization diagnostics is primarily a density-selection effect rather than ionization stratification.* The Orion data presented in this work do not establish this conclusion. The clean LVM density diagnostics in Orion do not span the high-ionization zones for which density stratification has been discussed most strongly. The traditional Orion diagnostics are principally [S II], [O II], and [Cl III]. The [Ar IV], C III], Si III], and N IV] measurements shown later in the manuscript come from external samples rather than from the same

Orion observations. Thus, the Orion experiment suggests that diagnostics with different atomic response functions can return different densities without requiring ionization stratification (though higher S/N data is needed to robustly prove the [S II], [O II], and [Cl III] densities are not in agreement), but it does not directly test whether the density differences between, for example, [S II] and [Ar IV] or C III] in extreme emission-line galaxies are produced primarily by this effect.

This distinction matters because the nebular environments in which strong ionization dependent density stratification has been reported are often very different from Orion. Extreme emission-line galaxies can differ simultaneously in ionization parameter, ionizing-spectrum hardness, metallicity and equilibrium temperature, pressure, compactness, star-formation surface density, density-bounded geometry, and the spatial filling factors of the  $S^+$ ,  $O^+$ ,  $O^{+2}$ ,  $Ar^{+3}$ , etc. zones. In such systems, ionization state and density may be physically correlated. A broad density PDF and ionization stratification are therefore not mutually exclusive hypotheses; they may be coupled manifestations of the same physical structure. So, the physically realistic picture could instead be that a regions pressure is a function of four correlated variables:  $n_e$ ,  $T_e$ ,  $\log U$ , and ionization and each line ratio samples a different projection of that multidimensional distribution. This framework would mean that  $n_M$  is not the sole organizing variable. When moving from [S II] to [Ar IV] to C III] or N IV], both atomic density sensitivity and physical ionization zone change. Those quantities are observationally correlated, so a correlation with  $n_M$  alone does not establish which physical variable is responsible.

**Requested revisions:**

1. Reframe the primary conclusions throughout the abstract, Results, and Discussion so that density-response effects are shown to be a potentially important contributor to diagnostic discrepancies rather than demonstrated to dominate over ionization stratification.
2. Clearly separate the empirical result established with Orion from the interpretation based on heterogeneous extragalactic samples.
3. Explicitly show which density diagnostics and ionization zones are actually available for Orion and acknowledge that the LVM measurements do not provide a direct test using the highest-ionization density tracers.
4. Investigate whether archival Orion UV measurements can provide an independent test. In particular, C III] and Si III] observations exist for multiple Orion positions (e.g. Rubin+1993). If aperture matching, signal-to-noise, or other limitations prevent their use, these limitations should at least be discussed.
5. For the extragalactic samples, test whether the inferred density correlates independently with  $n_M$  and ionization potential/excitation. At present these quantities are substantially confounded.

A particularly clean experiment would ultimately require simultaneous [S II], [O II], [Cl III], and [Ar IV], ideally supplemented by C III], Si III], and/or N IV], in the same systems, and an analysis of  $n_e$  versus  $n_M$  separately from  $n_e$  versus ionization state.

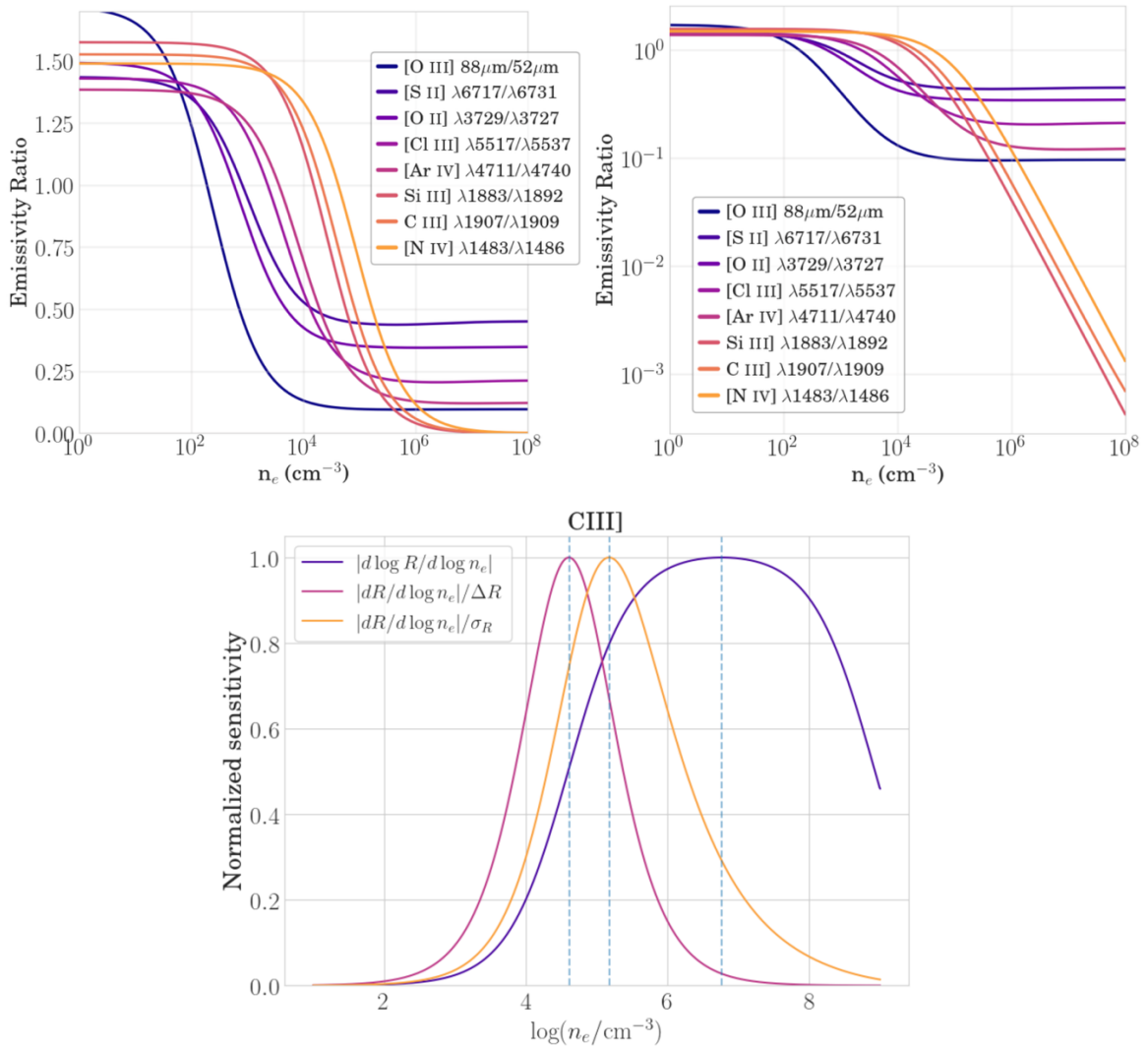
## 2. Physical interpretation of the maximum-sensitivity density $n_M$ is misleading.

- I. Because  $n_M$  is used as the central independent variable in several of the manuscript's principal figures, one of my largest concerns is that the current  $n_M$  definition does not correspond to maximum observational sensitivity. The manuscript defines  $n_M$  as the density at which  $|d \log R / d \log n_e|$  is maximized. While this is a mathematically well-defined measure of the maximum fractional response of a line ratio, I do not think it necessarily identifies the density at which the observable ratio has its greatest leverage, nor does it automatically identify the portion of an unresolved density distribution that contributes most strongly to the measured integrated ratio. This becomes particularly important for the UV doublet diagnostics which have large differences in their critical density ratios such as C III] 1907/1909, N IV] 1483/1487, and Si III] 1882/1892, for which the manuscript quotes  $\log n_M = 6.70, 7.14$ , and  $7.40$ , respectively. When the change in the observable ratio itself is examined, e.g.  $|dR/d \log n_e|$ , these diagnostics show their strongest absolute response at substantially lower densities, typically  $n_e \sim 10^4 - 10^5 \text{ cm}^{-3}$ . This difference is expected analytically. In the two-level approximation used by the authors,  $n_M = \sqrt{n_1 n_2}$ , whereas the maximum of  $|dR/d \log n_e|$  occurs near the lower critical density  $n_1 = n_M / \delta$ , where  $\delta = \sqrt{n_2 / n_1}$ . For diagnostics with large differences in their individual critical densities, the two scales can differ by several orders of magnitude. Indeed, the manuscript itself adopts  $\delta \simeq 200$  for C III], which implies  $n_M / \delta \sim 10^{4.5} \text{ cm}^{-3}$ , despite  $n_M \sim 10^{6.7} \text{ cm}^{-3}$ . Thus, the very large  $n_M$  values of these UV diagnostics are partly a consequence of defining sensitivity using a logarithmic derivative. A ratio can have a very large fractional response after the absolute ratio has already undergone most of its observable change.

This is important because much of the apparent dynamic range in Figs. 2 and 4 is supplied by these high- $n_M$  UV diagnostics. A large value of  $n_M$  does not necessarily imply that a diagnostic preferentially probes material near that density. There is also a useful internal consistency check in the authors' own forward model. The model is truncated at  $n_{\text{max}} = 10^6 \text{ cm}^{-3}$ , yet still produces meaningful inferred densities for C III], N IV], and Si III], whose quoted  $n_M$  values lie well above the maximum density present in the model. This demonstrates that these diagnostics can respond strongly to gas far below their quoted  $n_M$ .

More fundamentally, the quantity that determines how an unresolved density distribution affects an observed line is not simply  $|d \log R / d \log n_e|$ . For a density distribution  $H(n_e)$ , the luminosity contribution from each density interval depends on  $H(n_e) \epsilon_\lambda(n_e, T_e)$ . The numerator and denominator of the diagnostic can therefore have different density-dependent contribution functions. Those contribution functions seem much more directly related to the statement that a diagnostic “samples” a particular portion of a density PDF.

Sensitivities to the ratios discussed above can be seen in the plots below. The first row shows the diagnostic ratios vs  $\log n_e$  on the left and  $\log(\text{diagnostic ratios})$  vs  $\log n_e$  on the right. The right shows a large change in relative sensitivity at high densities for the UV lines, but for very small line ratios so the absolute change will be very small and unobservable. In the second row, the normalized sensitivity for different sensitivity metrics is plotted versus log density for CIII] as an example of a density diagnostic with a large difference in the critical densities of the individual lines. The purple curve reproduces the papers conceptual  $n_M$  definition that maximizes  $|d \log R / d \log n_e|$ . The pink curve shows where the actual measured ratio changes the most per dex in density. And the orange curve is the more observationally motivated maximum sensitivity (though only illustrative here as it uses Poisson statistics rather than a more rigorous version with an actual line flux uncertainty model).



**Requested revisions:** Because  $n_M$  underlies several central conclusions, I do not think clarification in the text alone is sufficient. I recommend that the authors:

1. Show  $R(n_e)$ ,  $|d \log R / d \log n_e|$ , and  $|dR / d \log n_e|$  for all the principal diagnostics, particularly C III], N IV], and Si III].

2. Define and show an observational-information metric that incorporates realistic uncertainties in the two measured line fluxes. For example, the relevant quantity could be related to  $|dR/d \log n_e|/\sigma_R$  rather than assuming constant fractional uncertainty in the ratio.
  3. Show the density-dependent emissivity/contribution functions  $H(n_e)\epsilon_{\text{numerator}}(n_e)$  and  $H(n_e)\epsilon_{\text{denominator}}(n_e)$  for representative diagnostics in the adopted forward model. This would directly demonstrate which densities dominate the integrated observables.
  4. Repeat the principal density-hierarchy plots using at least one alternative characteristic density derived from the linear-ratio response or observational-information metric.
  5. Report how the slope, significance, and inferred density-PDF parameters change under these alternative definitions.
  6. Clarify throughout that  $n_M$ , as currently defined, is specifically the density of maximum fractional ratio response and demonstrate that this quantity is an appropriate proxy for the physical density distribution sampled by each diagnostic.
- If the central trend remains comparably strong under these alternative definitions, that would significantly strengthen the manuscript.

### 3. The apparent $\sim 4$ -dex sensitivity baseline depends strongly on diagnostics with very different levels of robustness.

The manuscript emphasizes the large baseline in  $n_M$  covered by the Orion measurements. However, the baseline is much smaller if one considers only the robust, conventional, close-wavelength density doublets. These give approximately [SII]  $\log n_M = 3.06$ , [OII]  $\log n_M = 3.16$ , and [Cl III]  $\log n_M = 3.99$ , which is less than one dex in  $n_M$ . The much larger baseline requires two substantially more complicated classes of diagnostics: (i) transauroral/nebular combinations of [S II] and [O II], which have significant temperature sensitivity and very large wavelength baselines (making reddening uncertainties a concern) and (ii) Fe II/Fe III ratios involving a complex multilevel ion. The issue here is then that the current analysis gives similar statistical weight to diagnostics with substantially different physical and observational systematic uncertainties.

- I. Issues with Transauroral [S II] and [O II]: These ratios mix lines with very different excitation properties and widely separated wavelengths. Their inferred density therefore depends much more strongly on  $T_e$ , reddening correction, and relative spectrophotometric calibration than a close doublet such as [S II] 6717/6731. For [O II]  $\lambda\lambda 7320, 7330$ , recombination excitation may also contribute under some nebular conditions and should be addressed. The manuscript does account for temperature to some degree, but the quoted uncertainties should include the full propagation of the uncertainty in the adopted temperature, extinction correction, extinction law, and relative calibration for these widely separated lines.
- II. Issues with [Fe III]: I am also concerned by the treatment of the Fe diagnostics as multiple distinct density tracers. The authors explicitly acknowledge that several of these Fe diagnostics have multiple peaks (i.e.,  $\lambda 4658/\lambda 4702$ ,  $\lambda 4658/\lambda 4881$ , and  $\lambda 4702/\lambda 4881$ ) in their density sensitivity curves and so they choose the highest-

sensitivity peak lying above the minimum critical density to define  $n_M$ . This choice may be operationally useful, but I do not think it has yet been demonstrated that the resulting single  $n_M$  provides a robust description of the physical information in these ratios. Further, Laha+2017 show that  $\lambda 4881$  also has especially large uncertainties in the Einstein A-values. Only  $\lambda 4658/\lambda 4702$  doesn't use  $\lambda 4881$ , but Laha+2017 shows the sensitivity to density, where max sensitivity looks to be at  $10^5 \text{ cm}^{-3}$  on average, not  $10^7 \text{ cm}^{-3}$ . In author's Mendez-Delgado+2021 paper they show that most of the [Fe III] ratios are degenerate between multiple properties so are not strong density diagnostics but that "for  $10^3 \text{ cm}^{-3} < n_e < 10^6$ ,  $\lambda 4658/\lambda 4702$  (see bottom varies monotonically with  $n_e$  and is insensitive to  $T_e$ . Thus, it is the most reliable diagnostic in our case." But here you put the max sensitivity at  $10^7 \text{ cm}^{-3}$ , which is outside the window it is typically used (e.g., your previous work). This is contradictory. Mendez-Delgado+2021 also note that the shocks in Orion can cause Fe to be released from dust and can bias the Fe III emission even more strongly toward the dense shock front. This bias should be noted/discussed in the text. For these reasons, a simultaneous multilevel treatment of the [Fe III] lines (as done in the Esteban+2004 analysis of Orion using 14 [Fe III] lines or Mendez-Delgado+2021 using maximum-likelihood with  $\lambda 4658$ ,  $\lambda 4702$ ,  $\lambda 4734$ ,  $\lambda 4881$ ,  $\lambda 5271$ ) is more physically transparent than interpreting several pairwise ratios as independent characteristic density probes.

**Requested revisions:** I recommend a hierarchy of robustness tests.

1. Repeat the primary correlation analysis using:
  1. only conventional close-wavelength doublets;
  2. conventional doublets plus the transauroral [S II]/[O II] diagnostics;
  3. the above plus the Fe diagnostics.
2. For each case, report the slope, uncertainty, correlation coefficient/significance, and inferred density-PDF parameters. This would make clear which diagnostic classes are carrying the result.  
For the transauroral diagnostics, propagate uncertainties in  $T_e$ , reddening, extinction-law assumptions, and relative spectrophotometric calibration in addition to statistical line-flux uncertainties.
3. for Fe III, I strongly recommend presenting both:
  1. the individual-ratio analysis used here; and
  2. a simultaneous multilevel  $\text{Fe}^{+2}$  solution using all reliable detected lines, similar to the approaches used in Esteban+2004 and Méndez-Delgado+2021.

Please show whether a single  $T_e, n_e$  solution can reproduce the Fe III line intensities. If it cannot, the pattern of residuals itself may provide stronger evidence for unresolved density structure than treating the pairwise ratios as separate points. The manuscript should also explicitly distinguish algebraic independence from statistical/physical independence when ratios share measured lines and common atomic data.

**4. The forward model demonstrates that the proposed mechanism is sufficient, but it is not a discriminating test against plausible alternatives.**

The forward model intentionally imposes no ionization stratification and is nearly isothermal, specifically so that the hierarchy must arise from density-response functions. This is a perfectly sensible proof-of-concept experiment to address whether a broad density PDF can generate the Orion hierarchy even without ionization stratification (which is yes). But it does not answer how much of the observed [S II]–[Ar IV]–C III–N IV density hierarchy observed in extreme emission line galaxies comes from the density PDF, and how much comes from genuine ionization-dependent structure. A test of this requires a model in which both effects are allowed simultaneously and with conditions similar to EELGs.

I also think the manuscript needs a clearer physical description of what the proposed density PDF represents spatially. The current discussion can be read as suggesting that essentially the same broad density distribution exists throughout all ionization zones. It is not clear to me whether that is actually the intended physical picture or simply a simplifying assumption of the model. The manuscript itself later discusses the possibility that density inhomogeneities generate temperature structure through collisional de-excitation of important coolants. This makes the almost-isothermal assumption particularly important to distinguish as a numerical simplification rather than a physically self-consistent prediction.

**Requested revisions:** At minimum, the manuscript should explicitly characterize the current forward calculation as a proof-of-principle sufficiency test, rather than evidence that density-selection effects are the unique or dominant explanation.

A substantially stronger analysis would compare at least three model classes using the same observables:

1. a density-PDF-only model;
2. an ionization-stratified model;
3. a combined model in which density and ionization state are correlated.

This need not require a full radiation-hydrodynamic simulation. A parameterized multi-zone photoionization calculation in which  $n_e$ ,  $U$ ,  $T_e$ , and ionic fractions vary together would already provide a much more discriminating test. If such modeling is beyond the scope of the present paper, the conclusions should be narrowed accordingly.

**5. A physical picture is needed.**

The physical picture suggested by the paper seems to be one in which a density PDR exists uniformly throughout a nebula across all ionization zones. This physical idea of all densities existing in all ionization zones is very confusing. The proposed physical picture demonstrated with a model/simulation would be most useful. Are there simulations that exist that can demonstrate this? Can you construct your own? If not, a cartoon visual would be very useful to interpret the physical situation and what the density distribution looks like throughout the nebula: where the low- and high-density material resides

relative to the ionizing sources and how the different ionic zones intersect the density distribution. This would make it much easier to assess whether the proposed interpretation is physically consistent with the observations.

**6. The statistical treatment and uncertainty model require additional robustness tests.**

The apparent tightness of the density- $n_M$  relation is central to the interpretation, but I do not yet understand the uncertainty model sufficiently well to assess its significance.

In particular:

- the spread shown for the Orion measurements does not appear to represent the full measurement and systematic uncertainty associated with each diagnostic;
- different diagnostic classes clearly have different levels of systematic uncertainty;
- ratios that share emission lines and/or atomic data have correlated errors;
- the Bayesian density-PDF analysis assigns the same 0.25 dex uncertainty to all diagnostics, despite their substantially different observational and atomic uncertainties.

I do not think algebraic independence alone removes the covariance introduced by shared flux measurements and atomic data.

**Requested revisions:** I recommend that the authors:

1. Clearly distinguish the spatial dispersion among Orion spaxels from the uncertainty on each diagnostic density.
2. Explain and justify the adopted 0.25 dex uncertainty in the Bayesian modeling.
3. Repeat the Bayesian analysis using diagnostic-dependent uncertainties or plausible systematic uncertainty floors.
4. Include covariance among ratios that share emission lines wherever possible.
5. Perform a leave-one-diagnostic-out analysis to determine whether the fitted slope and PDF parameters are driven by one or two high-leverage diagnostics.
6. Repeat the fit excluding the UV diagnostics for which the interpretation of  $n_M$  is most strongly affected by the sensitivity-definition issue discussed above.
7. Report how the inferred slope and density-distribution parameters change under the diagnostic-subset tests recommended in Major Concern 3.

These analyses would help establish which aspects of the inferred density PDF are actually constrained by the observations and which depend on the adopted diagnostic and uncertainty model.

## Minor and specific comments

1. **High-redshift sample.** The high-redshift comparison appears extremely sparse and potentially non-representative. There are additional high- $z$  galaxies in the literature that satisfy the stated criteria. The sample should either be made as complete as reasonably possible or the high-redshift comparison should be explicitly presented as illustrative rather than evidence that the same relation applies across cosmic time.
2. **Additional density diagnostics.** Where possible, the authors should include additional independent constraints rather than restricting the comparison to the present set. In particular, the existing UV densities for Orion or the He I density-sensitive diagnostics may provide useful complementary information, although their temperature, optical-depth, and radiative-transfer dependencies would need to be modeled appropriately.
3. **Comparison with previous Orion measurements.** It would be useful to compare the inferred Orion densities directly with high-quality previous measurements, including the echelle analysis of Esteban+2004. Some of the densities in the present analysis appear substantially lower. The authors should discuss whether this arises from aperture/spatial averaging, line selection, atomic data, or another physical effect. As stated earlier, it would be useful to include a comparison to the Rubin UV densities. Additionally, Esteban+2004 report an [Ar IV] density. Please state why there is not [Ar IV] density reported for the LVM observations.
4. **Temperature structure and  $t^2$ .** The discussion suggesting that unresolved density structure may contribute to temperature inhomogeneities is interesting, but it is somewhat contradictory of the nearly isothermal forward model. The manuscript should distinguish clearly between what the model assumes and what the authors speculate may occur physically.
5. **Low-density limits in the forward-model figures.** I do not understand the physical interpretation of Figure 7 in which several diagnostics formally return densities far below the regime in which their ratios have appreciable sensitivity. A numerical inversion can return a density in the low-density asymptotic regime, but that does not imply that the density is observationally constrained. The authors should show the useful sensitivity range or low-/high-density limits for each diagnostic and distinguish a formal PyNeb inversion from a meaningful density measurement. This point is closely related to the  $n_M$  concern above.
6. **Terminology.** I suggest consistently distinguishing among “density-sensitive”, “maximum sensitivity density”, and “density preferentially sample” or similar versions of these terms. These concepts are currently treated as more interchangeable than I think is justified.